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Batch: 1
Experiment performed on (date): 18-09-2026

Title of the Experiment: Newton's Rings

Aim: To determine wavelength (λ) of light by Newton's rings experiment.

Apparatus: Travelling microscope, Sodium vapor lamp, Newton's ring apparatus

Observations:

Sr. No.	Ring No.	Travelling Microscope Reading (cm)						Diameter D_n (cm)=L+R	D^2_n	$D_m^2-D_n^2$
		On left = m+v (cm)			On Right = m+v (cm)					
		MSR	VSR	L	MSR	VSR	R			
1	8	0.1	0.047	0.147	0.1	0.047	0.147	0.294	0.086	0.080
2	7	0.1	0.038	0.138	0.1	0.038	0.138	0.276	0.076	0.070
3	6	0.1	0.026	0.126	0.1	0.026	0.126	0.252	0.064	0.058
4	5	0.1	0.016	0.116	0.1	0.016	0.116	0.232	0.054	0.048
5	4	0.1	0.010	0.110	0.1	0.010	0.110	0.220	0.048	0.042
6	3	0.05	0.035	0.085	0.05	0.035	0.085	0.170	0.029	0.023
7	2	0.05	0.016	0.066	0.05	0.016	0.066	0.122	0.014	0.011
8	1	0	0.038	0.038	0	0.038	0.038	0.076	0.006	

Calculations:

$$\text{Mean } (D_m^2 - D_n^2) = \frac{0.080 + 0.070 + 0.058 + 0.048 + 0.042 + 0.023 + 0.011}{7}$$

$$\therefore \text{Mean } (D_m^2 - D_n^2) = 0.047$$

$$\therefore \lambda = \frac{\text{Mean } (D_m^2 - D_n^2)}{4R} = \frac{0.047}{800} = 5875 \text{ \AA}$$

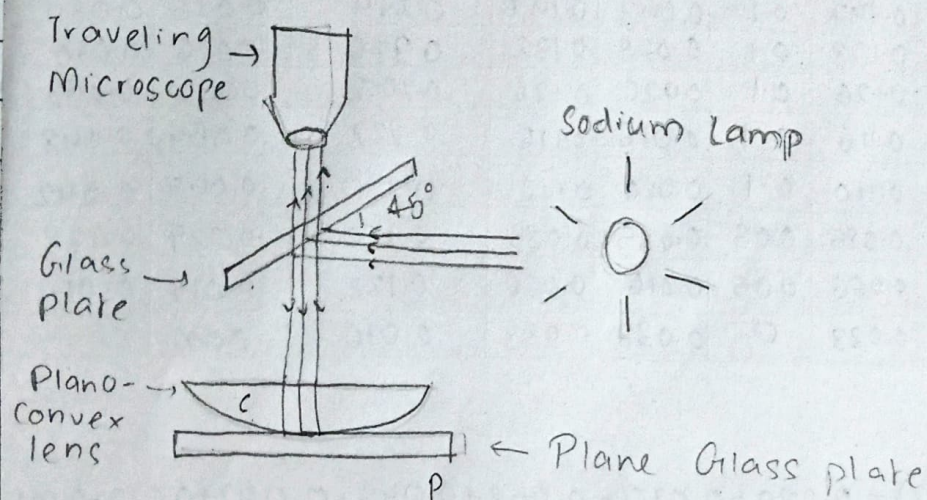


$$\text{Percentage Error} = \frac{5890 - 5875}{5890} \times 100 = \frac{1500}{5890} = 0.25\%$$

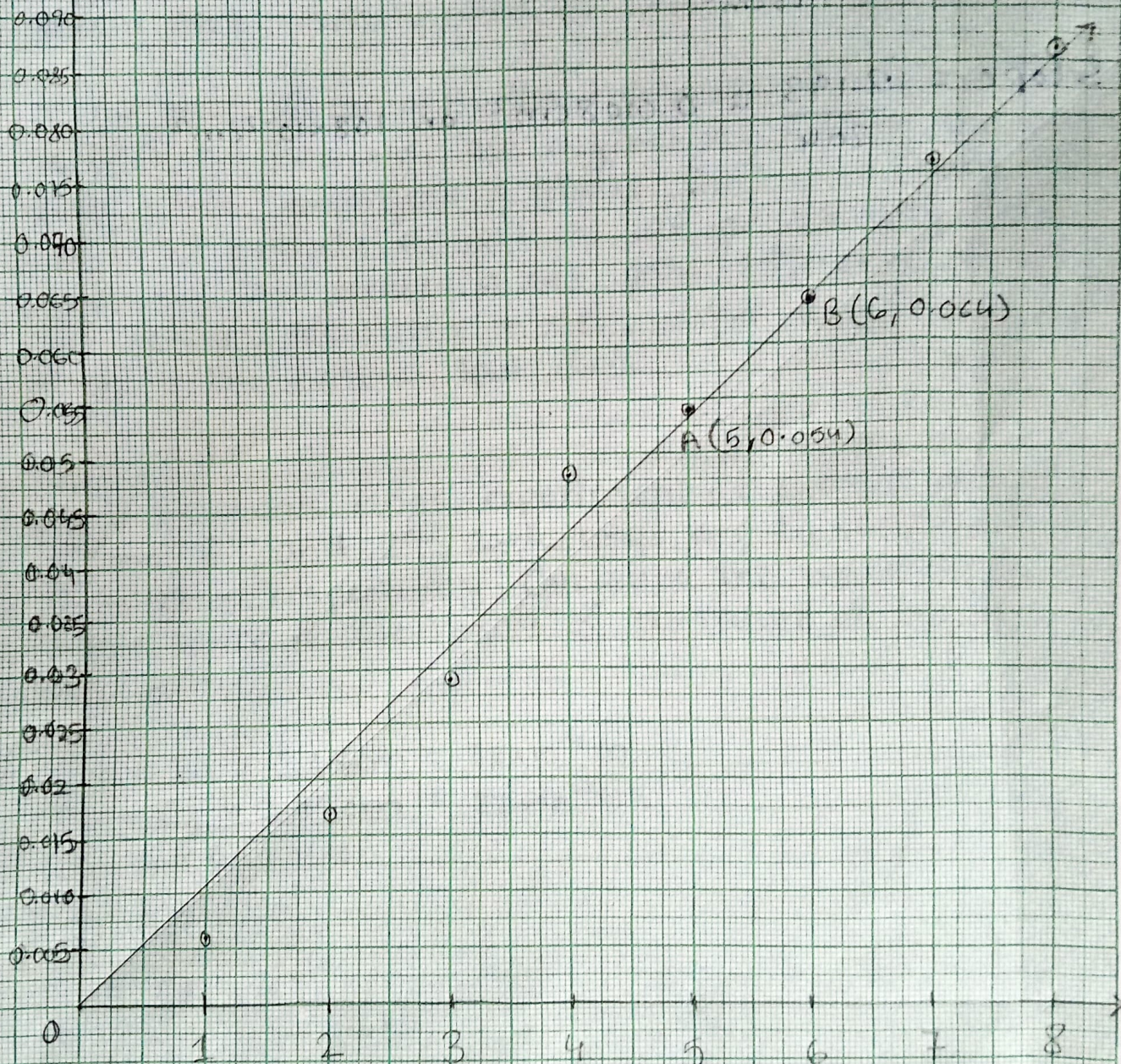
Result/s and Conclusion/s:

1. Wavelength (λ) of a given light is 5875 Å
2. Standard value of wavelength (λ) of a given light is 5890 Å
3. Percent error in obtained result 0.25%.

Diagram:



Square of diameter of rings $(D_m)^2$ $\rightarrow$



No. of rings (n) $\rightarrow$

On y-axis
1cm = 0.05

On x-axis
2cm = 1

$$\text{Slope} = \frac{\sum (n \cdot D_n^2)}{\sum n^2}$$

$$\sum n^2 = 1^2 + 2^2 + 3^2 + \dots + 8^2 = 204$$

$$\begin{aligned}\sum (n \cdot D_n^2) &= (1 \times 0.06) + (2 \times 0.017) + (3 \times 0.029) + \dots + (8 \times 0.086) \\ &= 2.193 \text{ cm}^2\end{aligned}$$

$$\text{Slope} = \frac{2.193}{204} \approx 0.0108 \text{ cm}^2 \text{ or } 1.08 \times 10^{-2} \text{ cm}^2$$

ASSIGNMENT!

Q.1. Why is the central spot of the Newton's rings interference pattern dark when viewed in reflected light?

→ At the point of contact, the thickness of the air film is zero ($t=0$). However, the ray reflected from the lower glass plate undergoes a phase shift of π (path difference of $\lambda/2$) because it reflects off an optically denser medium. This creates a net path difference of $\lambda/2$ at the centre, resulting in destructive interference and making the central spot dark.

Q.2. In a Newton's rings experiment, the measured diameters of the 6th and 2nd dark rings are $D_6 = 0.38\text{cm}$ and $D_2 = 0.22\text{cm}$ respectively. Calculate the wavelength of the light source if the radius of curvature of the lens is $R = 100\text{cm}$.

$$\rightarrow D_m = D_6 = 0.38\text{cm} = 38 \times 10^{-4}\text{m}$$

$$D_n = D_2 = 0.22\text{cm} = 22 \times 10^{-4}\text{m}$$

$$m - n = 6 - 2 = 4$$

$$\lambda = \frac{D_m^2 - D_n^2}{4(m-n)R}$$

$$= \frac{(38 \times 10^{-4})^2 - (22 \times 10^{-4})^2}{4(4)(100)} = \frac{960 \times 10^{-8}}{16} = 60 \times 10^{-8}\text{m} = 600\text{nm}$$

∴ The wavelength of the light source is 600nm.